

ALARM_SPEC - alarm clock for Matt's Timer

Fable's spec, 2026-09-27, base commit 7a8d5f1. Opus builds to this; Fable reviews the diff against it line by line. Tags: [measured] [design] [inferred] [verify].

1. Matt's words

2026-09-27, this repo's chat: "Okay lets add in an alarm clock feature next"

His answers to Fable's twelve questions, his words: "1. yes, 2. B, 3. B, 4. 12 hour, 5 A, 6. ignores mute but I need more sound options in needs to be something that can wake but not annoy me, 7. 15 minutes, 8. yes, 9. A, 10. B, 11. If it is verification testing just close it, 12. effort is high"

Decoded against the questions as asked:

#	Question	His answer	Means
1	Rings at a time of day, on a third tab ALARM	yes	third tab ALARM
2	How many alarms	B	several, each with its own on/off switch
3	Repeat	B	one-shot or chosen days of the week, per alarm
4	Time entry	12 hour	hour and minute wheels with the number pad, AM/PM
5	Snooze	A	no snooze, just STOP
6	Sound	see his words	the alarm ignores MUTE; more sound options, "something that can wake but not annoy me"
7	Ring length untouched	15 minutes	stops itself at 15 minutes, leaves a missed-alarm card
8	USE_EXACT_ALARM	yes	accepted
9	Survive a reboot, RECEIVE_BOOT_COMPLETED	A	accepted
10	Ring on a locked phone	B	full-screen takeover, USE_FULL_SCREEN_INTENT accepted
11	Base build	see his words	the open phone checks on f3ba666 are closed; build on it

2. Fable's design calls - not his words, his to overturn

Each of these is [design] by Fable. Opus builds them as written; Fable lists them to Matt in chat.

- D1. **It rings after a restart even before he unlocks the phone.** Android delivers `BOOT_COMPLETED` only after the first unlock. A phone that restarts overnight would stay silent until he picked it up. The alarm parts are therefore direct-boot aware and listen for `LOCKED_BOOT_COMPLETED` too (same permission, no new one).
- D2. **Late ring.** When the phone comes back up (or the app is updated, or the process returns) and an alarm's time passed 15 minutes ago or less, it rings now. Past more than 15 minutes, it does not ring: a missed-alarm card is posted instead.
- D3. **Volume ramp.** Gain rises in a straight line from 0.08 at the start of the ring to 1.0 at 90 seconds, then stays at 1.0. 1.0 is whatever his alarm volume already is; no stream volume is written.
- D4. **Sound choice is one setting for all alarms**, on the `ALARM` tab, separate from the timer's voice. Seven choices: four new wake voices (Section 7) plus `BELL`, `CHIME`, `PULSE`. Default is the first new voice.
- D5. **Vibration follows the existing VIB toggle** as read at the last time the app was open (mirrored into the alarm's own storage).
- D6. **Time zone and clock changes re-arm every alarm**, so 5:30 AM stays 5:30 AM local.
- D7. **No alarm labels, no per-alarm sound, no cap shown to him**; the list is capped at 20 alarms in code.

3. Walls

- No `INTERNET`. No dependency. No exported component besides the launcher activity: both new receivers, the new service and the new activity are `android:exported="false"`.
- The `<uses-permission>` lines are already in the manifest at the base commit, written by Fable. Opus adds components only and does not touch those lines.
- `TimerEngine.kt` and `TimerService.kt` are not changed by this job, except the one mirror call in D5 (`toggleVibe` and `init` call `AlarmEngine.mirrorVibe(vibe)`).
- No stream volume write, anywhere.
- Nothing he did not ask for.

4. Files

File	Owner	What
<code>AlarmEngine.kt</code>	Fable-owned	the alarm list, storage, next-trigger math, arming, fire handling, ring state
<code>AlarmService.kt</code>	Fable-owned	foreground service while ringing: wake lock, ring loop, notification, full-screen intent, missed card
<code>AlarmReceiver.kt</code>	Fable-owned	boot, locked boot, package replaced, time set, time zone changed: re-arm

File	Owner	What
AlarmActivity.kt	Opus	the full-screen ringing screen: time, STOP
MainActivity.kt, layouts	Opus	ALARM tab, list, editor overlay, sound button
Tones.kt	Opus	four wake voices, play returning a handle that can be stopped

All in android/app/src/main/java/com/matt/gymtimer/.

5. AlarmEngine (Fable-owned)

5.1 Storage

- `ctx.createDeviceProtectedStorageContext().getSharedPreferences("alarms", MODE_PRIVATE)`. Never the mt prefs: those are credential-protected and unreadable before the first unlock. AlarmEngine, AlarmService, AlarmReceiver and AlarmActivity must not call TimerEngine at all.
- Keys: list (String, JSON array via `org.json`, framework), nextId (Int), voice (Int, index into the alarm voice list), vibe (Boolean, D5), ringId (Int, -1 when not ringing), ringAtWall (Long, wall-clock ms the ring began).
- One alarm: `{"id":Int,"h":0..23,"m":0..59,"days":Int,"on":Boolean,"next":Long}`. days is a bitmask, bit 0 Sunday to bit 6 Saturday; 0 means one-shot. next is the wall-clock ms it is armed for, 0 when off.
- Written with `commit()` inside the fire path and the receiver (the process may die right after), `apply()` elsewhere.
- Malformed JSON or a bad field: that alarm is skipped, the rest load.

5.2 Next-trigger math

`nextTrigger(h, m, days, nowWall): Long`

- `Calendar.getInstance()` (default time zone), set to `nowWall`, then hour-of-day `h`, minute `m`, second 0, millisecond 0.
- If the result is `<= nowWall`, add one day.
- If `days != 0`, add a day at a time (at most 7) until the bit for that day of the week is set.
- Calendar stays lenient, so a time inside a daylight-saving gap lands on the next valid instant.

5.3 Arming

- `arm(a): AlarmManager.setAlarmClock(AlarmClockInfo(a.next, show), op)`.
- `op = PendingIntent.getForegroundService(ctx, a.id, Intent(ctx, AlarmService::class.java).setAction(ACT_FIRE).putExtra("id", a.id), FLAG_UPDATE_CURRENT or FLAG_IMMUTABLE)`.

- `show = PendingIntent.getActivity` to MainActivity, request code `100000 + a.id`, `FLAG_IMMUTABLE`.
- Wrapped in try/catch for `SecurityException` (API 31 and 32 without the permission); a failed arm sets `armFailed = true` for the UI.
- `disarm(a): AlarmManager.cancel(op)` built the same way, then `a.next = 0`.
- `armAll(nowWall):` for every alarm with on:
 - `next == 0` or `next > nowWall`: recompute next with `nextTrigger` and arm. (Recompute always: the time zone may have changed.)
 - `next <= nowWall` and `nowWall - next <= 15 min`: late ring (D2): start `AlarmService` with `ACT_FIRE` for it.
 - `next <= nowWall` and older: call `missed(a)`, then roll it: one-shot goes on = false, `next = 0`; repeating gets a new next and is armed.
- Public edits, each ending in `persist + listener callback`: `add(h, m, days)`, `update(id, h, m, days)`, `setOn(id, on)`, `delete(id)`, `setVoice(i)`, `mirrorVibe(v)`. add and update switch the alarm on. Turning on or saving always recomputes next from now.
- `init(ctx)` is idempotent, loads the list and calls `armAll`. Called from `MainActivity.onCreate`, `AlarmService.onCreate`, `AlarmReceiver.onReceive` and `AlarmActivity.onCreate`.

5.4 Fire

`fire(id)`, called by `AlarmService` on `ACT_FIRE`:

- 1 Alarm missing or `on == false`: return false (service stops itself if it is not already ringing).
- 2 next more than 60 s in the future: a stale intent; re-arm and return false.
- 3 Roll the alarm first, before any sound: one-shot goes on = false, `next = 0`; repeating gets `nextTrigger(h, m, days, now + 60 s)` and is armed. Persist with `commit()`.
- 4 If already ringing, keep the ring that is running (its clock is not reset) and return true.
- 5 Set `ringId`, `ringAtWall`, `ringFrom = elapsedRealtime()`, persist with `commit()`, return true.

`stop()`: clears `ringId` to -1, persists, tells the service to end the ring. Called by the STOP button on the screen and on the notification.

6. AlarmService (Fable-owned)

- Foreground, `specialUse`, not exported, not bound, direct-boot aware. `startForeground` is the first call in `onStartCommand`, every time, with `FOREGROUND_SERVICE_TYPE_SPECIAL_USE` on API 34 and up.
- Actions: `ACT_FIRE` (extra id), `ACT_STOP`. A null intent (sticky restart) resumes the ring if `ringId != -1` and the ring began 15 minutes ago or less by wall clock; otherwise posts the missed card and stops. Returns `START_STICKY`.
- Wake lock: partial, tag `matttimer:alarm`, not reference counted, acquired with a 16-minute timeout when the ring begins, released the moment the ring ends.
- Audio focus: `AUDIOFOCUS_GAIN_TRANSIENT` with `USAGE_ALARM` attributes when the ring begins, abandoned when it ends.

- Ring loop, main-looper Handler, one cue at a time with a generation counter (same pattern as `TimerEngine.postCue`):
- each fire reads the voice and vibe from `AlarmEngine`, plays the voice's phrase at gain $\min(1.0, 0.08 + 0.92 * t / 90 \text{ s})$ where `t` is `elapsedRealtime() - ringFrom`, and buzzes `longArrayOf(0, 300, 120, 300, 120, 500)` with `VibrationAttributes / AudioAttributes USAGE_ALARM` if vibe;
- the next fire is posted at phrase length + 1500 ms;
- no phrase starts at `t > 15 min (900 000 ms)`. When that cut is reached the ring ends as MISSED.
- The alarm never reads `TimerEngine.sound`. MUTE does not reach it.
- Ending the ring (STOP or the 15-minute cut): kill the cues, stop the playing track now (Section 7, the handle), cancel the vibration, abandon focus, release the wake lock, clear `ringId`, `stopForeground(STOP_FOREGROUND_REMOVE)`, `stopSelf()`. On the 15-minute cut, post the missed card first.
- Channels, both created in `onCreate` with sound null and vibration off (the service makes every sound):
- `alarm_ring, IMPORTANCE_HIGH, VISIBILITY_PUBLIC`;
- `alarm_missed, IMPORTANCE_DEFAULT`.
- Ringing notification (id 8): ongoing, category `CATEGORY_ALARM`, title `ALARM`, text the alarm's time in 12-hour form (5:30 AM), one action `STOP` (`PendingIntent.getService, ACT_STOP`), content intent and `setFullScreenIntent(pi, true)` both to `AlarmActivity`, `FOREGROUND_SERVICE_IMMEDIATE` on API 31 and up.
- Missed card (id 9 + alarm id): not ongoing, auto-cancel, title `MISSED ALARM`, text the alarm's time, content intent to `MainActivity`.
- `onTaskRemoved` does nothing.

7. Tones (Opus)

- `play(pcm, gain)` returns a handle (the `AudioTrack` or a small wrapper) with `stop()`, so `STOP` silences a phrase mid-ring. Existing callers ignore the return value; their behavior is unchanged.
- Four new wake voices, his words: "something that can wake but not annoy me". Constraints [design]: every fundamental at or below 900 Hz; attack 40 ms or longer (the current 14 ms stays for the existing cues); phrase 4 to 8 seconds; no clipped pure tones; each clearly different from the others in contour and tempo. Names are one short word each, Opus's choice. Each gets a preview of 3 seconds or less.
- The alarm voice list is the four new ones followed by `BELL`, `CHIME`, `PULSE` (their chime phrase). The timer's voice cycle stays `BELL`, `CHIME`, `PULSE`, `MUTE` and does not gain the new ones.
- Phrases are rendered on first use, not at startup, so the timer's startup cost does not grow.

8. AlarmReceiver (Fable-owned)

- Not exported, direct-boot aware. Intent filter: `BOOT_COMPLETED`, `LOCKED_BOOT_COMPLETED`, `MY_PACKAGE_REPLACED`, `TIME_SET`, `TIMEZONE_CHANGED`.
- `onReceive`: `AlarmEngine.init(ctx)` then `AlarmEngine.armAll(now)`. Nothing else. No work off the main thread is needed; the list is at most 20 entries.

- [verify] on the phone: that a non-exported receiver gets the system's boot broadcast. If the build or the platform refuses it, that is a BLOCKED, not an exported receiver.

9. Manifest (Opus adds components only)

- `<service android:name=".AlarmService" android:exported="false" android:directBootAware="true" android:foregroundServiceType="specialUse">` with the `PROPERTY_SPECIAL_USE_FGS_SUBTYPE` property: "Local alarm clock that must ring with the app closed and the phone locked".
- `<receiver android:name=".AlarmReceiver" android:exported="false" android:directBootAware="true">` with the filter in Section 8.
- `<activity android:name=".AlarmActivity" android:exported="false" android:directBootAware="true" android:showWhenLocked="true" android:turnScreenOn="true" android:launchMode="singleInstance" android:excludeFromRecents="true" android:taskAffinity="">`.
- MainActivity, TimerService and the `<application>` element are not made direct-boot aware.

10. Screens (Opus)

10.1 ALARM tab

- Third tab button ALARM after STOPWATCH, portrait and landscape. A tap on it goes through the existing onTab (so it also silences a timer ring-out, as the other tabs do).
- In this mode the presets, the stage, the progress bar, the laps and the three big buttons are hidden. Shown instead:
 - a status line: NEXT 5:30 AM · IN 7H 12M, or NO ALARM SET;
 - the list, one row per alarm, sorted by time of day: the time large in 12-hour form, under it ONCE or the days (EVERY DAY, WEEKDAYS, WEEKENDS, or M W F style), and an on/off control at the right with a touch target of 56 dp or more. Tapping the row opens the editor; tapping the control only switches it;
 - ADD ALARM button, full width, same height as the big buttons;
 - the sound button, showing the current alarm voice name; a tap cycles to the next voice and plays its preview at full gain.
- Warning lines, shown only when true, in amber:
 - notifications not granted: NOTIFICATIONS OFF · TAP TO ALLOW (requests POST_NOTIFICATIONS; asked by itself the first time an alarm is saved);
 - API 34 and up and `NotificationManager.canUseFullScreenIntent()` is false: FULL SCREEN OFF · TAP TO ALLOW, opening `Settings.ACTION_MANAGE_APP_USE_FULL_SCREEN_INTENT` for this package;
 - `AlarmEngine.armFailed`: ALARM COULD NOT BE SET.
- The status line's countdown refreshes once a minute while the tab is showing, and on every engine callback.

10.2 Editor overlay

- Same look as the timer's picker. Hour wheel 1 to 12, minute wheel 00 to 59 (formatter, not displayedValues - see the typed-5 finding in README.md), both typeable with the number pad. An AM / PM pair of buttons, one lit.
- Seven day buttons S M T W T F S, each a toggle. None lit means it rings once.
- CANCEL, SAVE, and DELETE (editing an existing alarm only). The buttons stay above the keyboard the same way the picker's do. The number pad's check key commits the typed value and drops the keyboard; it does not save.
- A new alarm opens on 6:00 AM, no days. Back closes the editor.
- Survives rotation the way the picker does.
- After SAVE a short line in the status area reads RINGS IN 7H 12M.

10.3 AlarmActivity

- Full screen over the lock screen, screen on, kept on while ringing. Black background, the alarm's time large, ALARM above it, one STOP button spanning the width, at least 96 dp tall.
- STOP calls AlarmEngine.stop() and finishes the activity. Back does nothing while ringing. When the ring ends any other way the activity finishes itself.
- Touches AlarmEngine only.

11. Docs (Opus, same commit as the behavior)

README.md gains an **Alarm** section and the permission paragraph reads eight permissions; tools/make_pdf.py changes with it and docs/TIMER.pdf is re-run. Everything on-phone is tagged [verify, n=0].

12. Checks that prove it

Build: android\gradlew.bat -p android assembleRelease passes on the committed tree. Greps on the source: no INTERNET, no setStreamVolume, adjustVolume or adjustStreamVolume; no TimerEngine reference in the four alarm files; exported="true" appears once in the manifest.

Phone checks, Fable's, each on Matt's word: alarm 2 minutes out rings with the app closed and the screen off; full screen over the lock screen; STOP from the screen and from the card; MUTE on the timer and the alarm still sounds; ramp heard by him; each voice previewed by him; one-shot switches itself off; repeating alarm shows the next day; 15-minute cut and missed card; restart the phone with an alarm set and do not unlock; time zone change; timer running while the alarm fires.